

RESULTS the MØLLER POLARIMETER STUDIES

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WHAT's NEW?

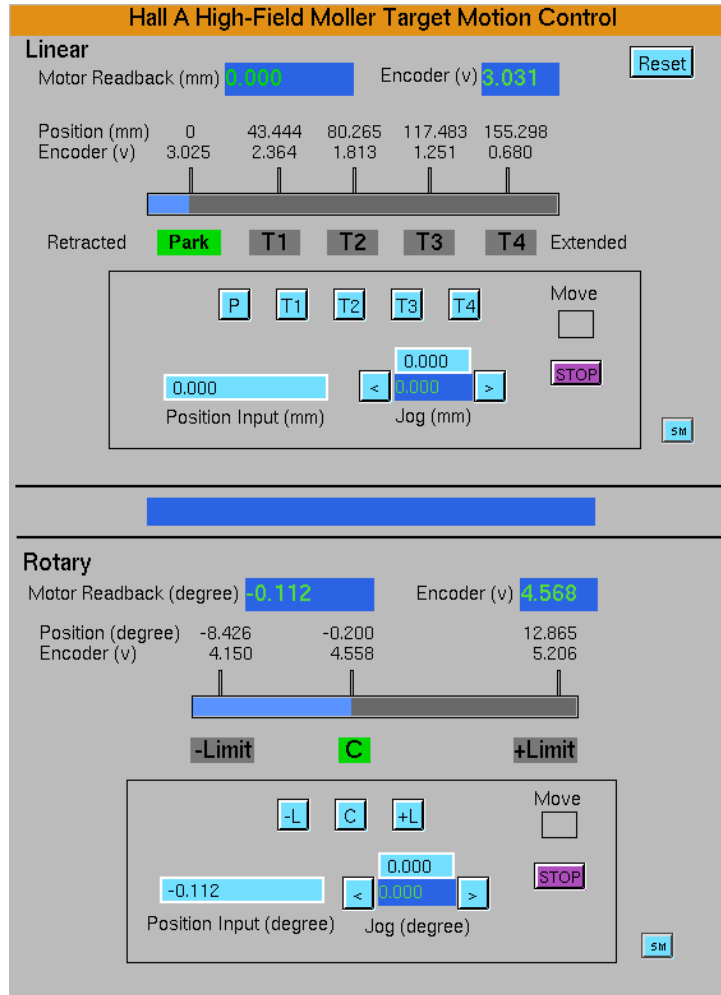
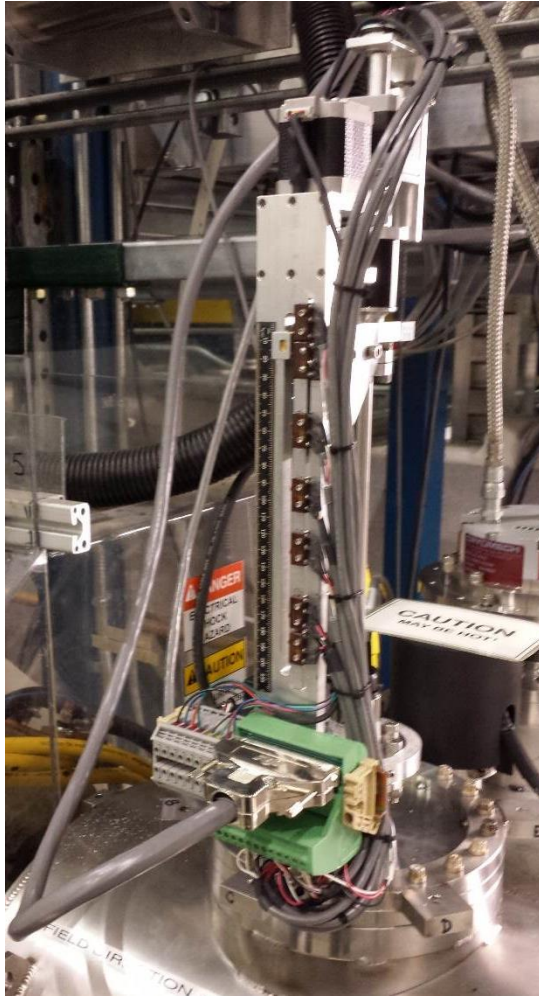
New SC magnet

New target

Five measurements + 1 test:

- *18-20 December, 2015 -11GeV*
- *17 February, 2016 - 4.48GeV*
- *19 February, 2016 - beam charge asymmetry test, 4.48GeV*
- *29 February, 2016 – spin dance, 8.82GeV*
- *31 March, 2016 – 10.99GeV*
- *19 April, 2016 – with Compton chicane, 10.99GeV*

NEW POLARIZED ELECTRON TARGET



Designed and made by Temple University

Rotation in horizontal plane $\pm 10^\circ$

Vertical translation

Targets in holder: pure iron $1\mu\text{m}$, $4\mu\text{m}$, $12\mu\text{m}$, $25\mu\text{m}$

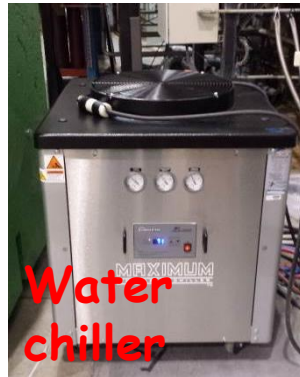
NEW MAGNET



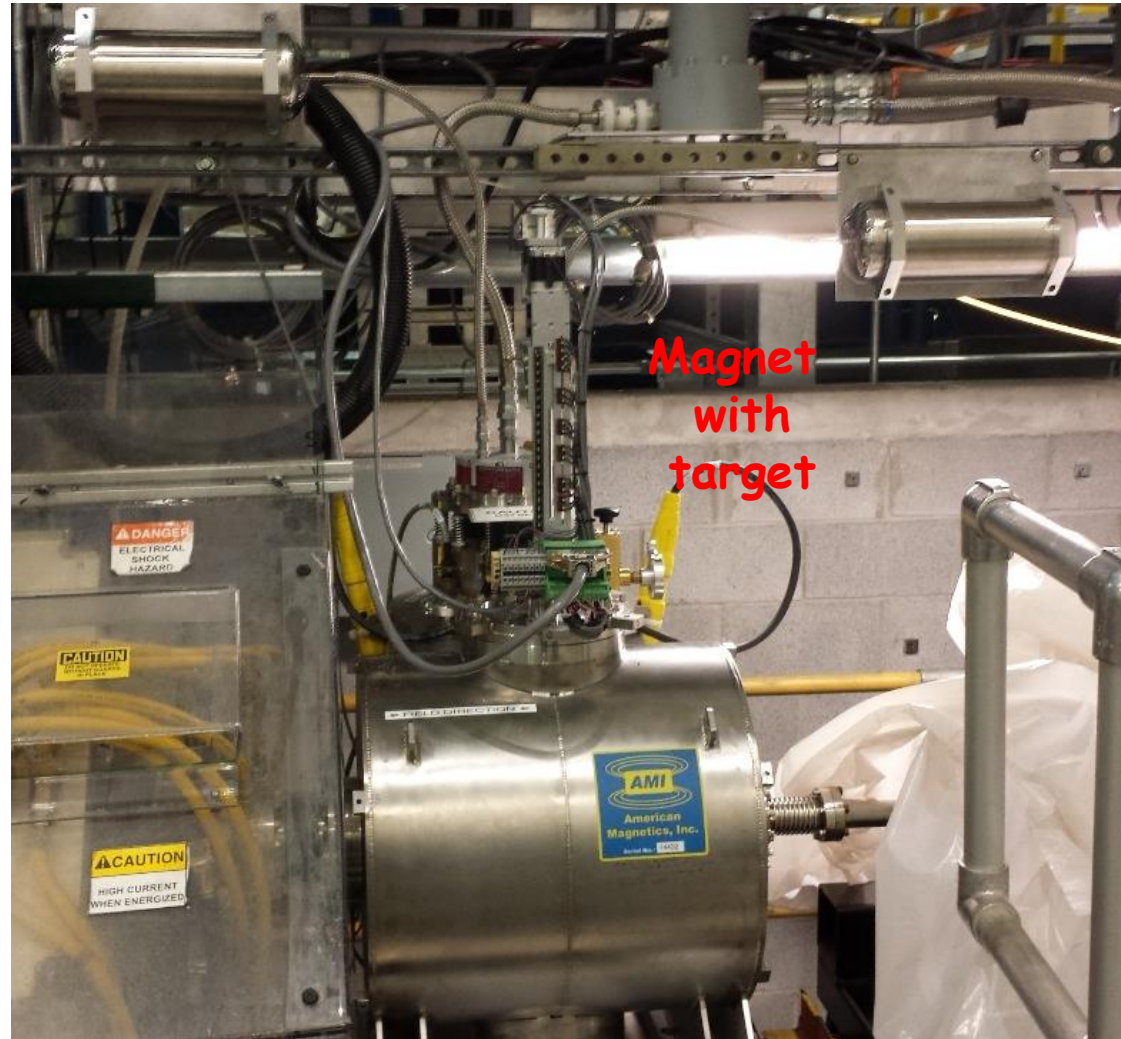
Power supply



Compressor



Water chiller



Magnet with target

New superconductive magnet:

Maximal field $\pm 5T$

- portable

- no LN required

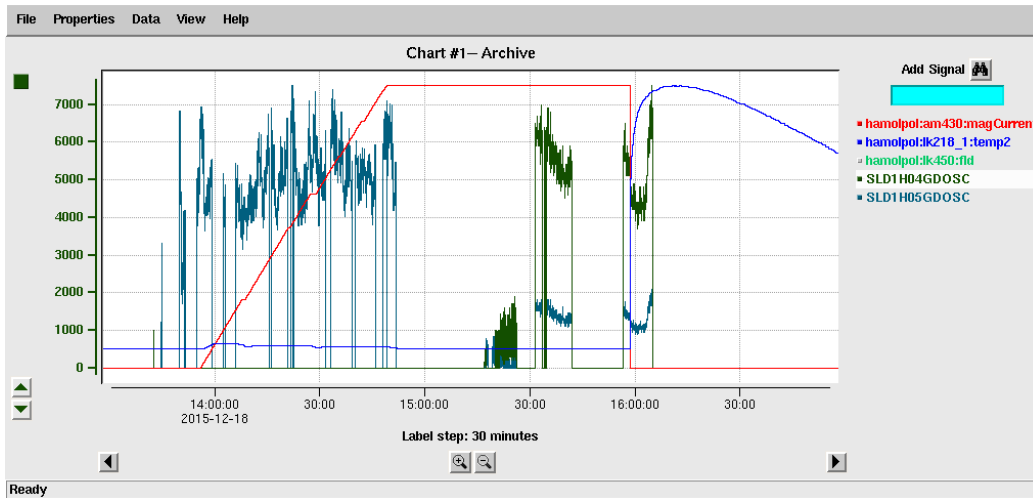
- no cryo lines required

Small amount of He $\rightarrow$

- low thermal capacity and “cold” production $\rightarrow$

- slow field ramp-rate ($\sim 35\text{min: } 0 \rightarrow 3T$)

FIRST EXPERIENCE WITH OPERATION



18 December, 2015

CW beam 1.5μA

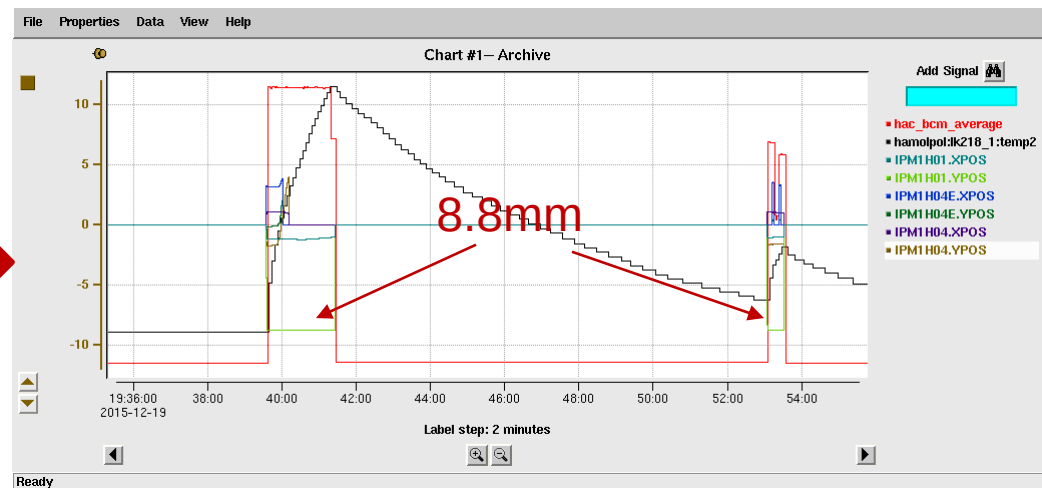
4T field in magnet

Target #3 (12μm)

Magnet QUENCH -

5-6 hours to recovery

19 December, 2015
Beam through Compton
Pulse/Tune
Low radiation
No field in magnet
No Moller target



EXPERIENCE WITH MAGNET OPERATION

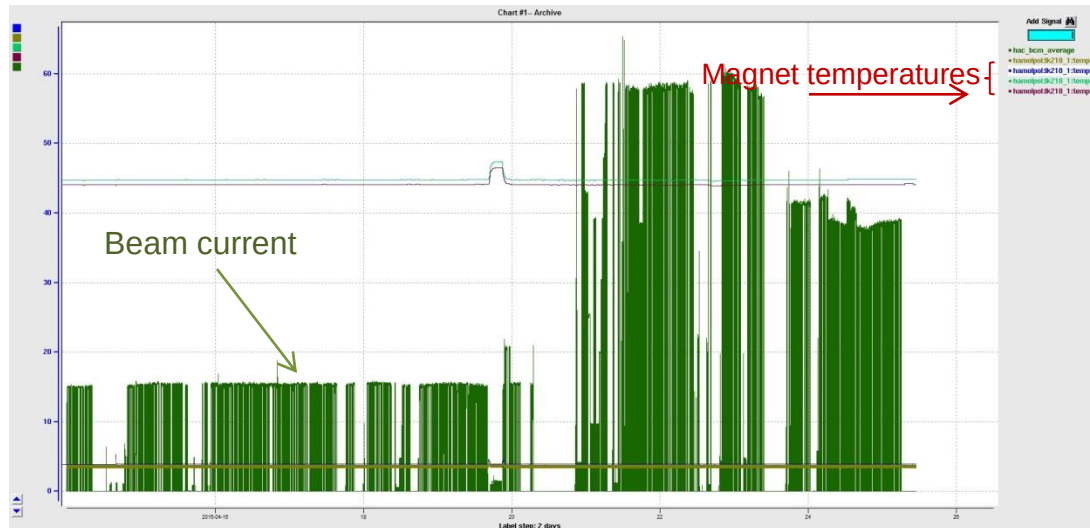
Magnet can be used at high beam current and high radiation in Hall A

04/19/2016 – First Moller measurement with beam through the Compton chicane (11GeV)

High radiation:

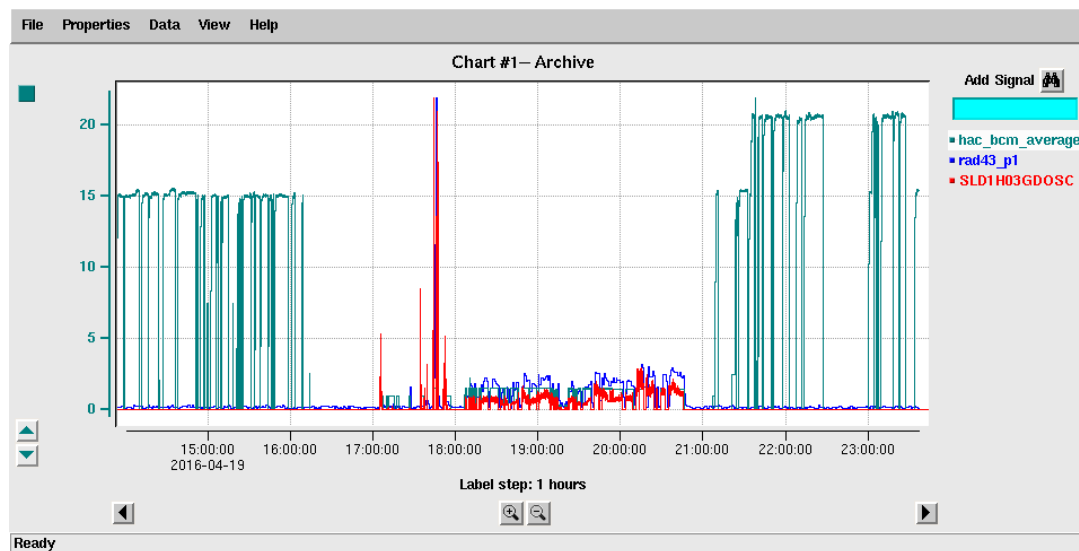
- *Hall A target in “EMPTY” not at “HOME”;*
 - *Optics combined with Compton;*
 - *Combination of all above.*
- Will be checked in the fall run.*

Magnet alignment test at low beam energy is needed.



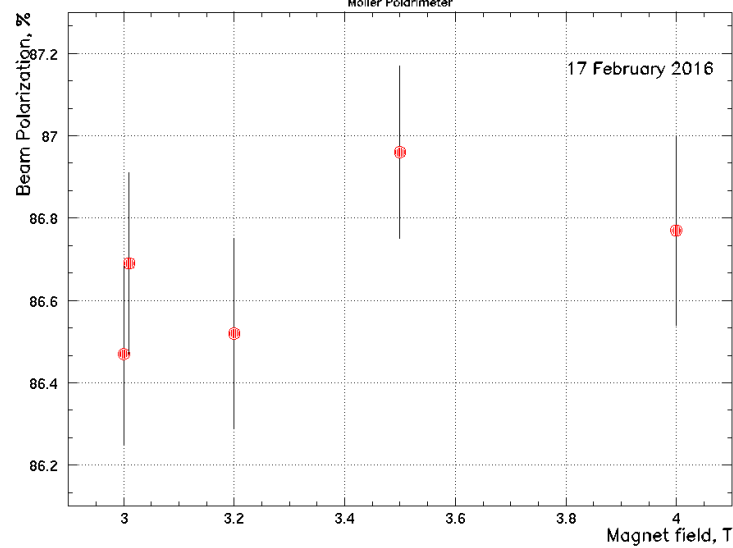
Rad43_p1 – Hall A track ramp (up to 0.9 mrem/h)

SLD1H03GDOSC – target ion chamber (up to 50 rad/h)

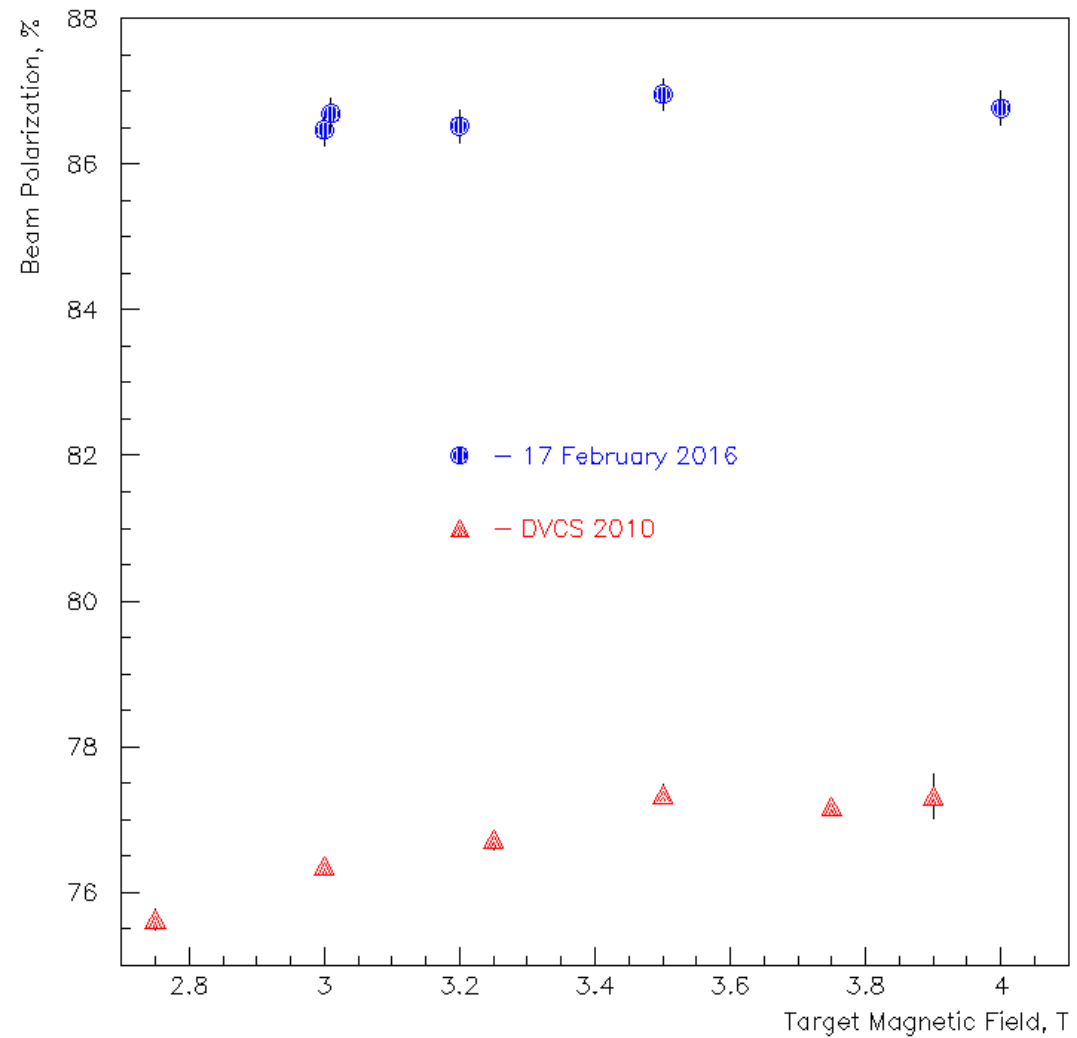


NEW TARGET SATURATION TEST

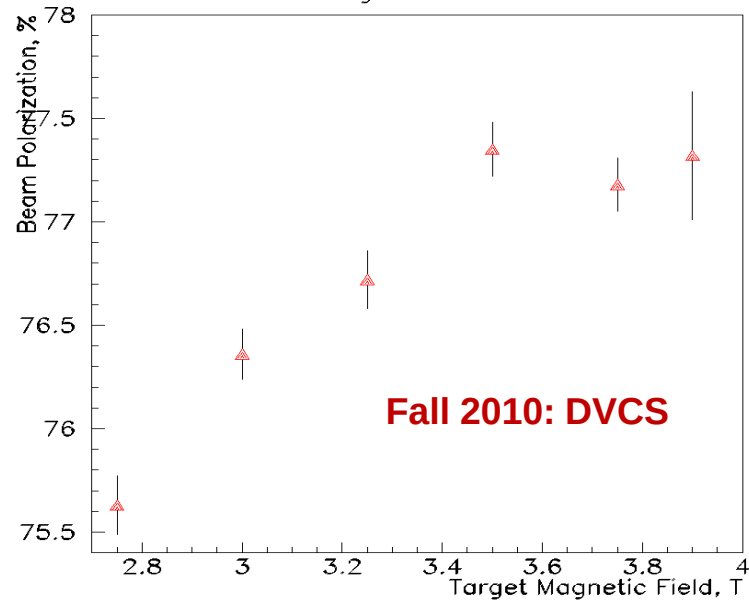
Measured Polarization vs. Target Saturation Field



DVCS Target Saturation Test



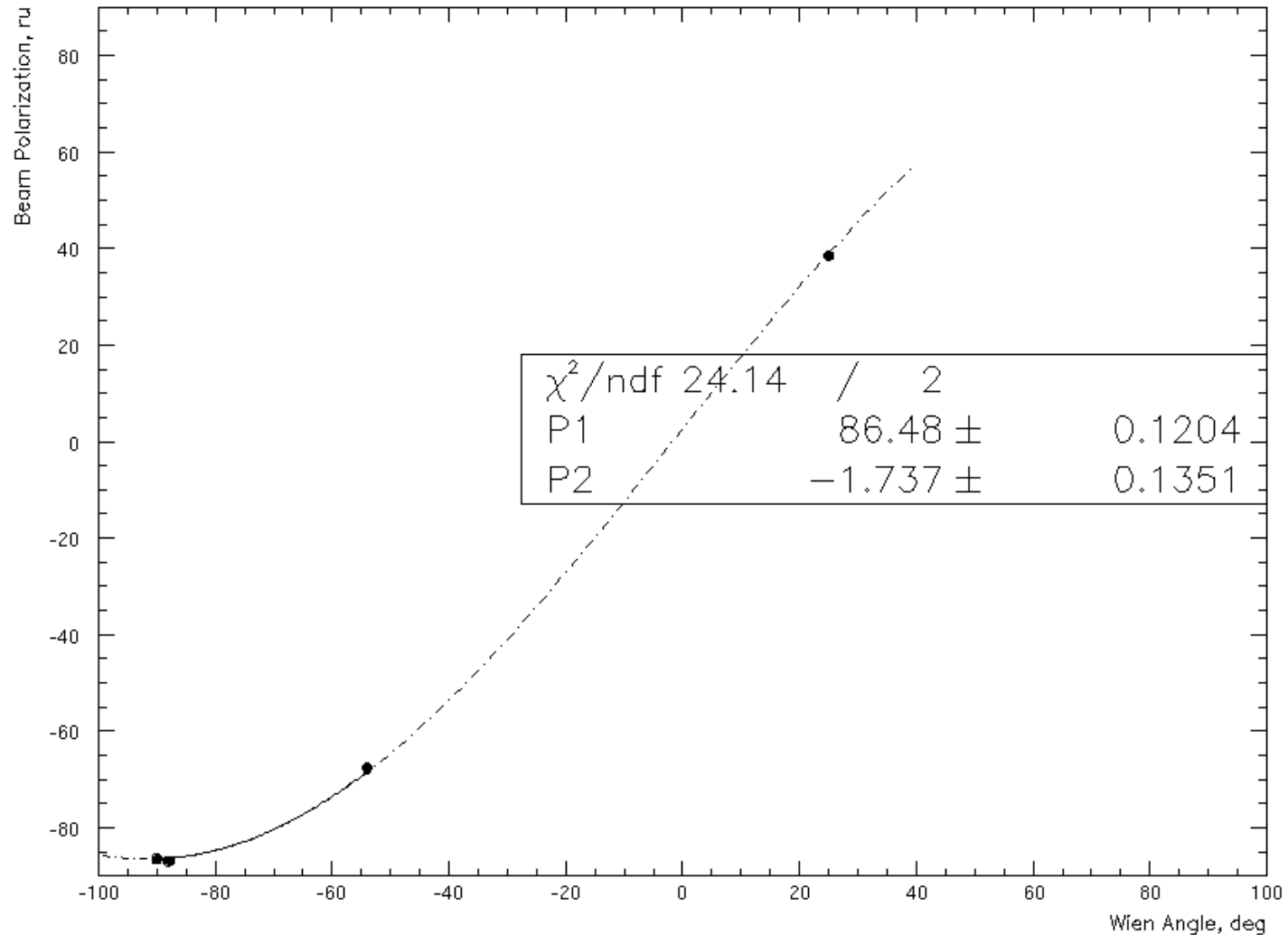
DVCS Target Saturation Test



SPIN DANCE

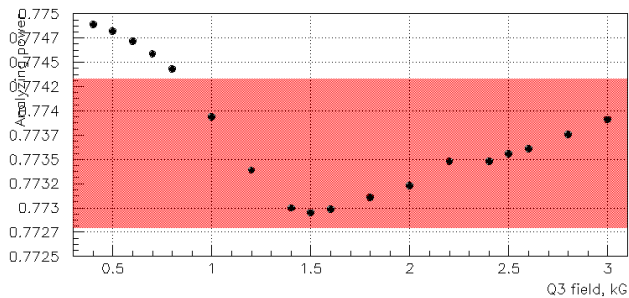
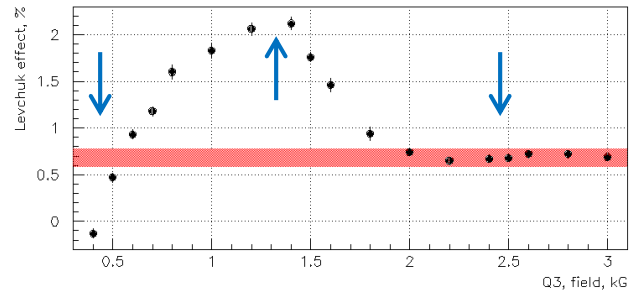
Moller results for Spin Dance 02/29/2016

Moeller run=16009

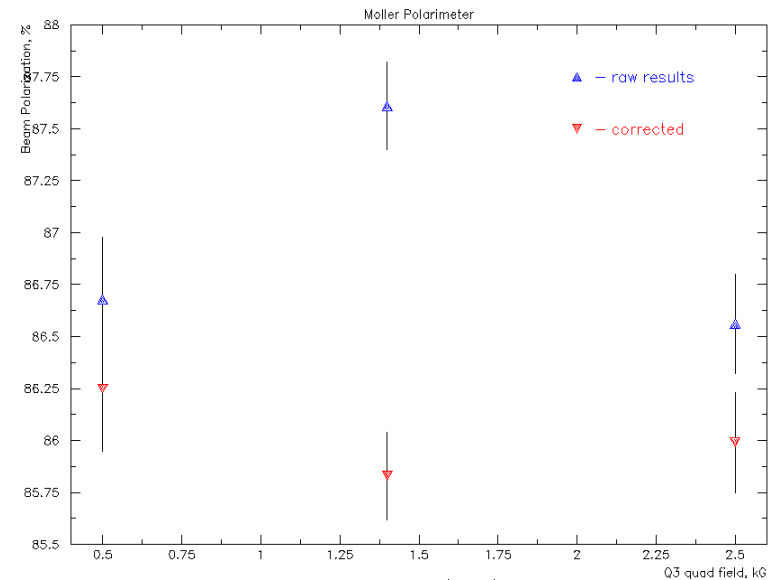


GEANT vs. MEASUREMENT SYSTEMATICS

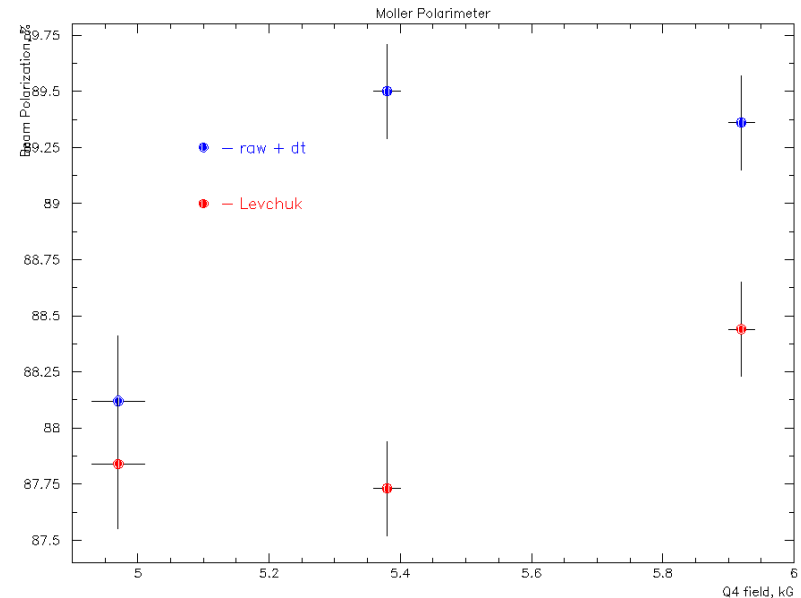
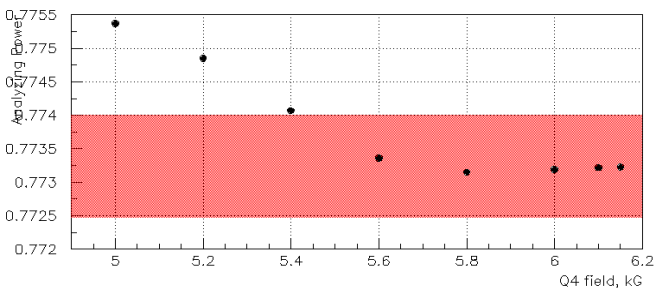
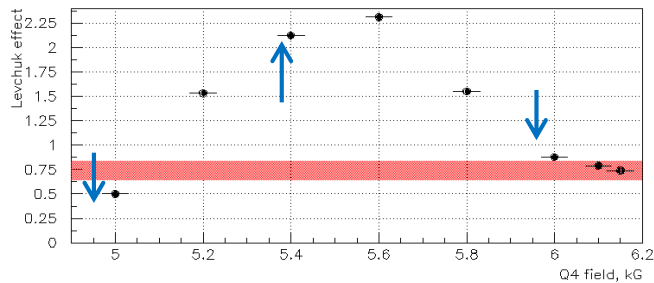
Moller results, 03/31/2016



$E_b = 11\text{GeV}$



Moller results, 04/19/2016



MORE SYSTEMATICS

Dead time:

- can estimate it with using LED pulser*

To do list:

- Target improvement;*
- Background (GEANT);*
- Rad. corrections;*
- Fe vs. Co and Ni;*
- Kerr effect to study target alignment.*